AKOSET INJECTION 2ML AMP

DESCRIPTION: AKOSET INJECTION is a colourless or almost colourless solution

COMPOSITION: Each ampoule contains Frusemide 20 mg/2 ml

PHARMACODYNAMICS: Frusemide has been demonstrated to inhibit primarily the reabsorption of sodium not only in the proximal and distal tubules but also in the loop of Henle. The high degree of efficacy is largely due to this unique site of action. The action on the distal tubule is independent of any inhibitory effect on carbonic anhydrase and aldosterone. The diuresis following an intravenous injection commences within five minutes, and somewhat later after intramuscular injection. The diuresis reaches its peak within the first half hour and lasts for two hours. Frusemide is often effective in patients with markedly reduced glomerular filtration rates in whom other diuretics usually fail. This is because frusemide inhibits the reabsorption of a very high percentage of the filtered sodium. Frusemide has been used concurrently with aldosterone antagonists to good advantage.

PHARMACOKINETICS: Absorption and Fate: Frusemide is incompletely but fairly rapidly absorbed from the gastrointestinal tract: bioavailability has been reported to be about 60 to 70% but is reduced in renal failure. It has a biphasic half-life in the plasma with a terminal elimination phase that has been estimated to range up to about 1½ hours although it is prolonged in renal and hepatic insufficiency. It is up to 99% bound to plasma proteins and is mainly excreted in the urine, largely unchanged. Variable amounts are also excreted in the bile, non-renal elimination being considerably increased in renal failure. Frusemide crosses the placental barrier and is excreted in milk.

Additional data for IV administration of frusemide are as follow:

i) Protein binding - very high (91 to 97%; almost totally to albumin)
ii) Half-life - wide variation among individuals normal: ½ to 1 hour
iii) Onset of action - intravenous: 5 minutes
iv) Time to peak effect - intravenous: within 30 minutes
v) Duration of action - intravenous: 2 hours
vi) Elimination - renal (88%)
- faecal / biliary (12%)

INDICATION: For the treatment of the oedema associated with congestive heart failure, cirrhosis of the liver and renal disease, including the nephrotic syndrome. Particularly useful when an agent with greater diuretic potential than that of the more commonly used agents is desired.
Intravenous administration is indicated when a rapid onset of diuresis is desired, eg. acute pulmonary oedema. Intramuscular or intravenous route is indicated if gastrointestinal absorption is impaired or oral medication is not practicable for any reason.
Note: Parenteral administration should be reserved for patients with whom oral medication is not practical.

RECOMMENDED DOSAGE:
Adults:Parenteral administration: Parenteral administration of furosemide should be reserved for patients in whom oral medication is not recommended. Parenteral therapy should be replaced with oral therapy, as soon as this is practical, for the continued mobilisation of oedema. The usual initial dose of furosemide is 20 to 40 mg given as a single dose, injected intramuscularly or intravenously. The intravenous injection should be given slowly (1 to 2 minutes). Ordinarily a prompt diuresis ensues. Depending on the patients response a second dose can be administered two hours after the first dose, or later.

If the diuretic response with a single dose of 20 to 40 mg is not satisfactory: e.g. In a patient refractory to maximal doses of thiazides, the following schedule should be used under careful medical supervision: increase this dose by increments of 20 mg, not sooner than two hours after the previous dose, until the desired diuretic effect has been obtained. This individually determined single dose should then be given once or twice daily.

Acute pulmonary oedema: Since the diuresis evoked by furosemide given intravenously commences within five minutes and leads to an intensive diuresis, the use of furosemide intravenously in the treatment of patients with acute pulmonary oedema has proven particularly valuable. The following schedule is recommended: 40 mg of furosemide is immediately injected slowly, intravenously followed by another 40 mg dose. 1 to 1½ hours later if that is indicated by the patients condition. If deemed necessary additional therapy (e.g. digitalis, oxygen) can be administered concomitantly.

Infants and children:
Paediatric administration: Parenteral administration of furosemide should be reserved for patients in whom oral medication is not recommended. Parenteral therapy should be replaced by oral therapy as soon as this is practical, for continued mobilisation of oedema. For maintenance therapy in infants and children the dose should be adjusted to the minimum effective level. The usual initial dose of furosemide injection (intravenously or intramuscularly) in infants and children are 1 mg/kg body weight and should be given under close medical supervision. If the diuretic response after the initial dose is not satisfactory, dosage may be increased by 1 mg/kg not sooner than two hours after the previous dose, until the desired effect has been obtained. Doses greater than 6 mg/kg body weight are not recommended.

ROUTE OF ADMINISTRATION: i.m. or slow i.v. injection

CONTRAINDICATIONS: Frusemide is contra-indicated in women of childbearing potential because animal reproductive studies have shown that it may cause foetal abnormalities. Exceptions to the above are life threatening situations where the use of a diuretic such as frusemide is specially indicated as opposed to the use of alternative drugs.
The physician of course, should balance this efficacy potential against the teratogenic and embryotoxic potential demonstrated to occur in animal studies.
Frusemide is contraindicated in anuria. If increasing azotaemia and oliguria occur during treatment of severe progressive renal disease, the drug should be discontinued. In hepatic coma and in states of electrolyte depletion, therapy should not be instituted until the basic condition is improved or corrected. Frusemide is also contraindicated in patients with a history of hypersensitivity to this compound.

WARNINGS & PRECAUTIONS: As with any potent diuretic electrolyte depletion may occur during therapy with Frusemide, especially in patients receiving higher doses and a restricted salt intake. Electrolyte depletion may manifest itself by weakness, dizziness, lethargy, leg cramps, anorexia, and vomiting and/or mental confusion. In oedematous hypertensive agents being treated with antihypertensive agents, care should be taken to reduce the dose of these drugs when frusemide is administered since frusemide potentiates the hypotensive effect of antihypertensive medications.
Asymptomatic hyperuricemia can occur and gout may rarely be precipitated. Reversible elevations of BUN may be seen. These have been observed in association with dehydration, which should be avoided, particularly in patients with renal insufficiency.
Cases of tinnitus and reversible hearing impairment have been reported. There have also been some reports of cases in which irreversible hearing impairment occurred. Usually, ototoxicity has been reported when frusemide was injected rapidly in patients with severe impairment of renal function at doses exceeding several times the usual recommended dose and in whom other drugs known to be ototoxic were often given. If the physician elects to use high dose parenteral therapy in patients with severely impaired renal function, controlled intravenous infusion is advisable (for adults it has been reported that an infusion rate not exceeding 4 mg frusemide per minute has been used).
Periodic checks on urine and blood glucose should be made in diabetics and even those suspected of latent diabetes when receiving frusemide. Increases in blood glucose and alterations in glucose tolerance tests with abnormalities of the fasting and two-hour postprandial sugar has been observed and rare cases of precipitation of diabetes mellitus have been reported.
Frusemide may lower serum calcium levels and rare cases of tetany have been reported. Accordingly, periodic serum calcium levels should be obtained.
Patients receiving high doses of salicylates, as in rheumatic diseases in conjunction with frusemide may experience salicylate toxicity at lower doses because of competitive renal excretory sites.

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It has been reported in the literature that diuretics such as frusemide may enhance the nephrotoxicity of cephaloridine. Therefore, frusemide and cephaloridine should not be administered simultaneously. Sulphonamide diuretics have been reported to decrease arterial responsiveness to pressor amines and to enhance the effect of tubocurarine. Great caution should be exercised in administering curare or its derivatives to patients undergoing therapy with frusemide, and it is advisable to discontinue frusemide two days prior to any elective surgery.

INTERACTION WITH OTHER MEDICAMENTS:

1. Alcohol or hypotension-producing drug when used concurrently with frusemide may potentiate the hypotensive or diuretic effect.
2. Used concurrently with amiodarone may lead to an increased risk of arrhythmias associated with hypokalemia.
3. Amphotericin B (parenteral). There is a potential increase of ototoxicity and nephrotoxicity when used concurrently or sequentially and especially when renal function impairment is present. In addition electrolyte imbalance may be intensified.
4. ACE inhibitors. Concurrent use with frusemide may cause sudden and severe hypotension particularly so in sodium and volume-depleted patients.
5. Anticoagulants when used with frusemide may decrease anticoagulant effects due to reduction of plasma volume. Diuretic-induced improvement of hepatic-congestion may result in increased procoagulant factor synthesis and hence dosage adjustment may be necessary.
6. Frusemide when used with Insulin may rarely raise blood glucose concentration or interfere with the hypoglycemic effects. Adjustment of hypoglycemic medication may be necessary.
7. Used with nonsteroidal anti-inflammatory drugs (NSAIDs) especially Indomethacine may antagonist natriuresis and increase in plasma renin activity. Indomethacin and other NSAIDs with the exception of diflunisal may also reduce the increase in urine volume caused by loop diuretics possibly by inhibiting renal prostaglandin synthesis and/or by causing sodium and fluid retention. Inhibition of renal prostaglandin synthesis caused by concomitant use with NSAIDs may increase risk of renal failure secondary due to a decrease in renal blood flow.
8. Hypokalemia-causing medications. Used with these drugs increased risk of severe hypokalemia may be encountered, serum potassium concentration and cardiac function monitoring and potassium supplementation may be required.
9. Lithium. It is not recommended to be used concurrently with loop diuretics unless close monitoring of the patient is possible because of the reduced renal clearance which may provoke lithium toxicity.
10. Nephrotoxic and ototoxic medications. Especially in the presence of renal function impairment, the potential for ototoxicity and nephrotoxicity may be increased, the concurrent and/or sequential use of loop diuretics should be avoided.
11. Chloral hydrate. Administration of chloral hydrate followed by intravenous frusemide may result in diaphoresis, hot flashes and variable blood pressure including hypertension due to a hypermetabolic state cause by displacement of thyroxine from its bound state.
12. Probenecid. Concurrent use with probenecid has been found to increase serum concentration frusemide by inhibiting active renal tubular secretion.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Furosemide crosses the placental barrier and should not be given during pregnancy unless there are compelling medical reasons. It should only be used for the pathological causes of oedema which are not directly or indirectly linked to the pregnancy. The treatment with diuretics of oedema and hypertension caused by pregnancy is undesirable because placental perfusion can be reduced, so, if used, monitoring of fetal growth is required. However, furosemide has been given after the first trimester of pregnancy for oedema, hypertension and toxaemia of pregnancy without causing fetal or newborn adverse effects.

Use in Lactation: Breast-feeding

Furosemide is contraindicated as it passes into breast milk and may inhibit lactation.

SIDE EFFECTS: Various forms of dermatitis, including urticaria and rare cases of the exfoliative dermatitis, erythema multiforme, pruritus, paraesthesia, blurring of vision, postural hypotension, nausea, vomiting, or diarrhoea, may occur. Anaemia, leukopenia, aplastic anaemia and thrombocytopenia (with purpura) may occur. Rare cases of agranulocytosis have occurred which have responded to treatment. Cases of tinnitus and reversible hearing impairment have been reported. There have also been some reports of cases in which irreversible hearing impairment occurred. Usually, ototoxicity has been reported when frusemide was injected rapidly in patients with severe impairment of renal function at doses exceeding several times the usually recommended dose and in whom other drugs known to be ototoxic were often given.

In addition to the following rare adverse reactions have been reported: however, relationship to the drug has not been established with certainty: sweet taste, oral and gastric burning, paradoxical swelling, headache, jaundice, thrombophlebitis and emboli and acute pancreatitis. In children, complaints of mild to moderate abdominal pain and cramping have been reported after intravenous frusemide.

Frusemide-induced diuresis may be accompanied by weakness, fatigue, light-headedness or dizziness, muscle cramps, thirst, increased perspiration, urinary bladder spasm, and symptoms of urinary frequency. As far as hyperglycaemia is concerned, see Precautions.

Transient pain after intramuscular injection has been reported at the injection site.

By far the most frequently encountered problem is excessive depletion of blood volume, which can lead to profound shock, frequently complicated by hypokalaemia, and ending in death.

Effects on Ability to Drive and Use Machine

Reduced mental alertness, dizziness and blurred vision have been reported. Particularly at the start of treatment, with dose changes and in combination with alcohol. Patients should be advised that if affected, they should not drive, operate machinery or take part in activities where these effects could put themselves or others at risk.

SYMPTOMS AND TREATMENT OF OVERDOSE: Over dosage will lead to fluid and salt depletion subsequent to initial excessive diuresis. Medical help must be called in without delay so as to ensure immediate measures for an adequate restoration of water and electrolyte balances

INCOMPATIBILITIES: Furosemide may precipitate solutions of low pH, and therefore dextrose solutions are not suitable infusion fluids for furosemide injection. The injection solution should not be mixed with other drugs in infusion bottles.

STORAGE CONDITIONS: Store below 30°C. Protect from light.

[Note: This product is light sensitive and should remain in carton until ready for use. Do not use if solution is yellow].

KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.

SHELF LIFE: Please refer to outer package.

PACK SIZES: Pack in 10's x 2ml ampoule in a box.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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